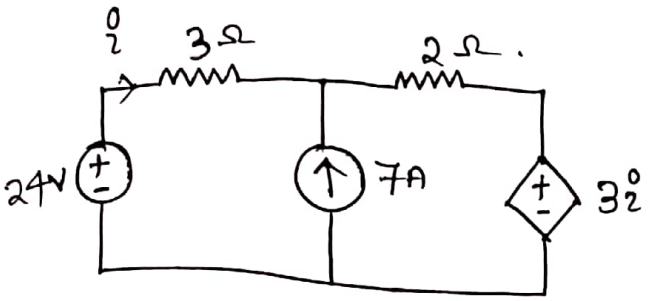
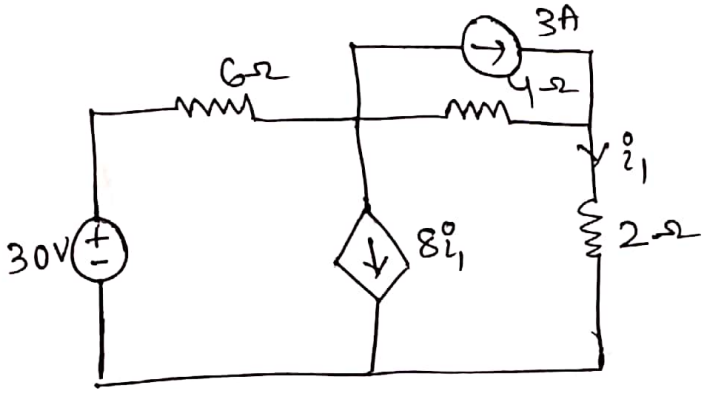
ET Assignment 2

1)



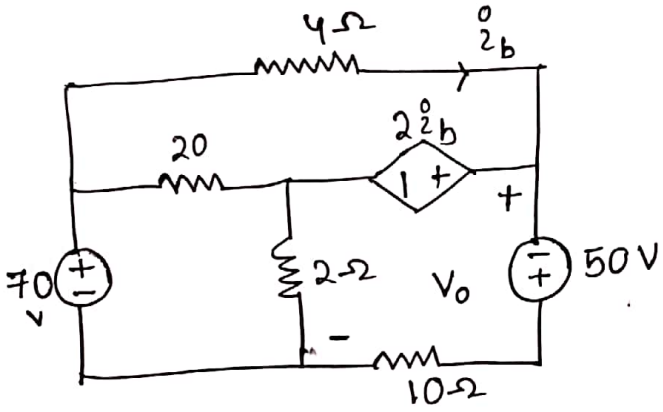
Solve the current 'i' using superposition theorem.

2)



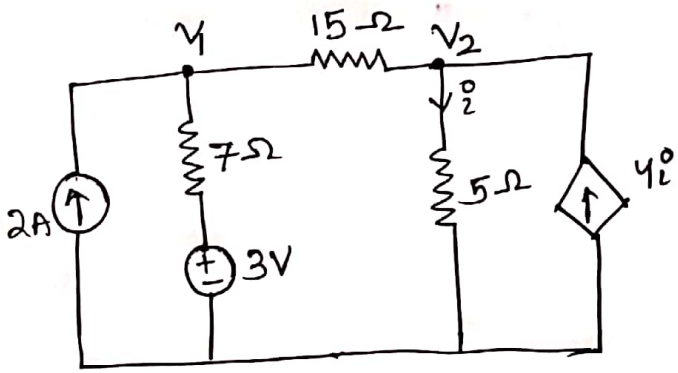
Solve the current 'i_1' using superposition theorem.

3)



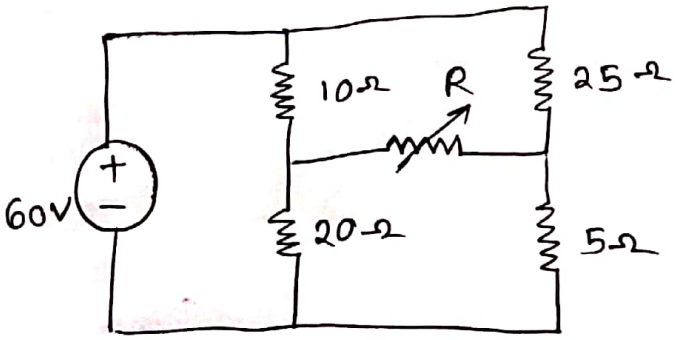
Solve the voltage V_o using superposition theorem.

4)



Solve for the voltages V_1 & V_2 across the current sources.

5)



Determine the maximum power that can be delivered to the variable resistor R.

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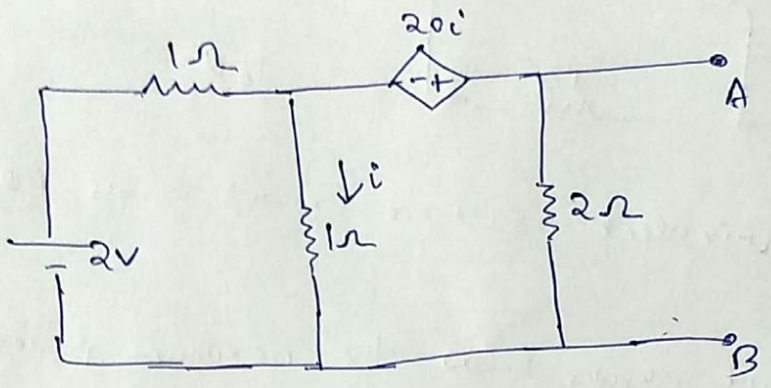
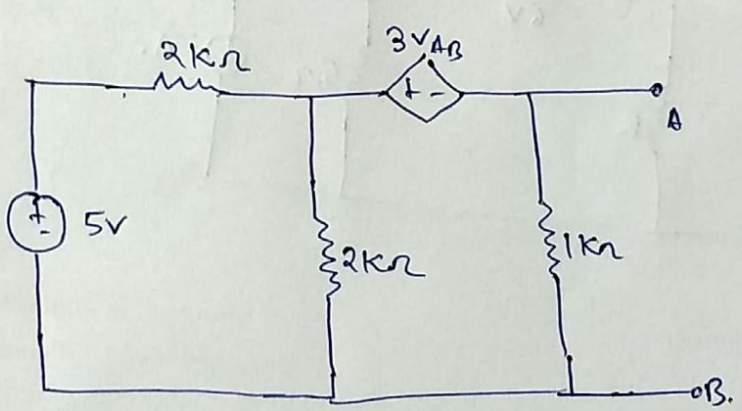
Assisment . 2

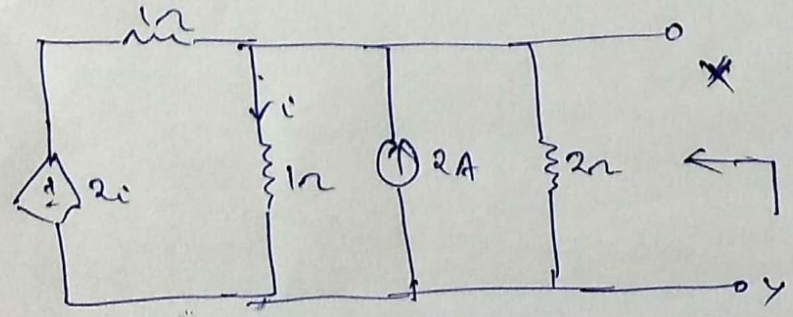
Q6 - For the given circuit, the Thevenin equivalent is to be determined. The Thevenin voltage, V_{Th} (in volt) seen from terminal AB is ____ ?



Q7 - For the circuit given below what is the Thevenin's voltage and resistence across the terminals A & B.



Q8 - For the circuit shown in the figure, the Thevenin voltage and resistence looking into x-y are



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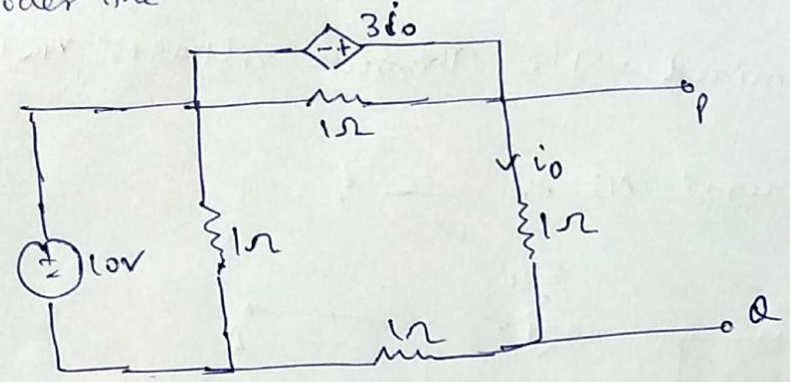
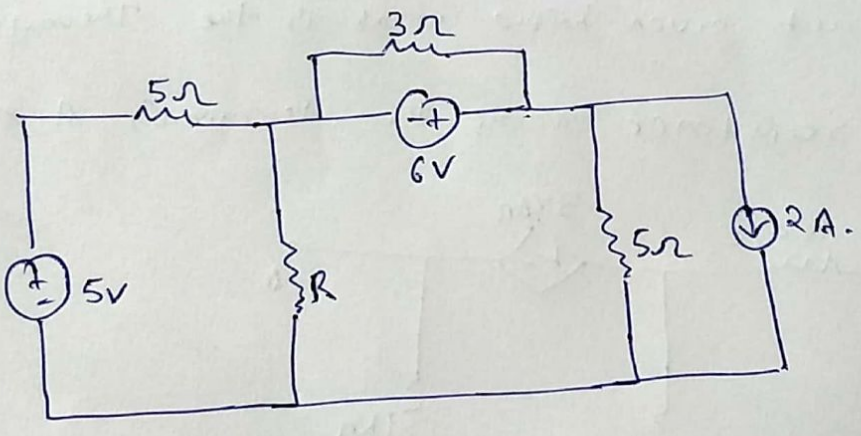
Q9. Consider the circuit shown in figure



The Thevenin's equivalent resistence (in Ω) across PQ is ___

Q10. In the circuit shown below the maximum power transferred to the resistor 'R' is ___ W.



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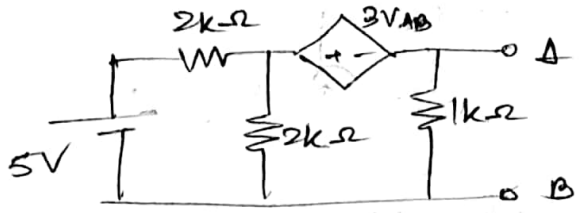
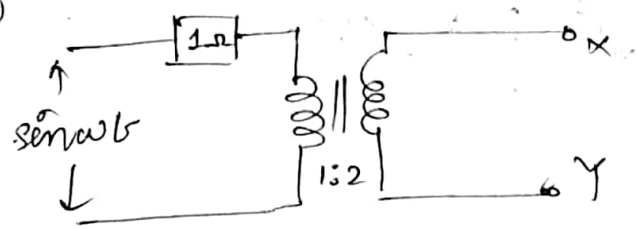
(11) Find the Norton's Resistance across A & B :

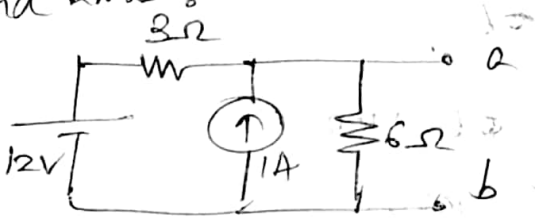


(12)

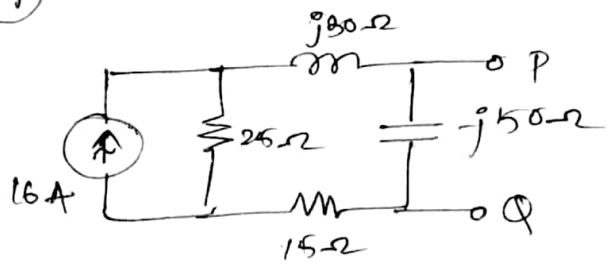


Find the Norton's ~~Re~~ Impedance across X-Y and the short circuit current through X-Y.

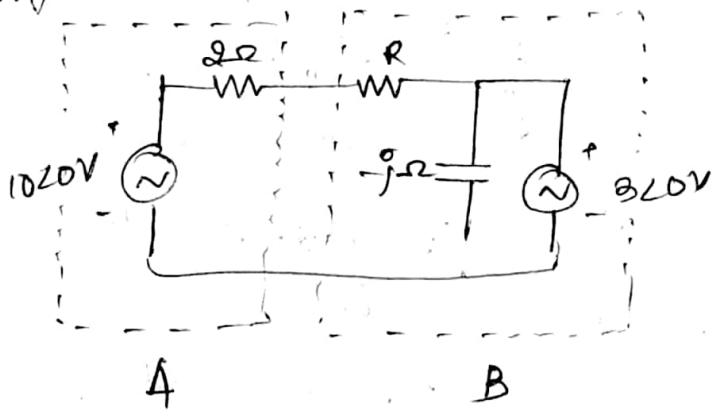
(13) Find the Norton's equivalent current across ab ? and $R_{N_{ab}}$?



(14) Find Norton's equivalent current through P & Q.



(15) Find value of R for which maximum power is transferred from ckt A to B ?



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